

MTH 1035 Handout – Abstract Vector Spaces

Definition of an abstract vector space (taken from Elementary Linear Algebra by Anton and Rorres).

Let V be an arbitrary nonempty set of objects on which two operations are defined: addition, and multiplication by scalars. By addition we mean a rule for associating with each pair of objects $\mathbf{u}$ and $\mathbf{v}$ in V an object, called the sum of $\mathbf{u}$ and $\mathbf{v}$; by scalar multiplication we mean a rule for associating with each scalar k and each object $\mathbf{u}$ in V an object $k\mathbf{u}$, called the scalar multiple of $\mathbf{u}$ by k . If the following axioms are satisfied by all objects $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V and all scalars k and m , then we call V a vector space and we call the objects in V vectors.

Rule 1 If $\mathbf{u}$ and $\mathbf{v}$ are objects in V , then $\mathbf{u} + \mathbf{v}$ is also in V .

Rule 2 $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

Rule 3 $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$

Rule 4 There is an object $\vec{0}$ in V , called a zero vector for V , such that

$$\vec{0} + \mathbf{u} = \mathbf{u} + \vec{0} = \mathbf{u} \quad \text{for all } \mathbf{u} \text{ in } V.$$

Rule 5 For each $\mathbf{u}$ in V , there is an object $-\mathbf{u}$ such that

$$\mathbf{u} + (-\mathbf{u}) = (-\mathbf{u}) + \mathbf{u} = \vec{0}$$

Rule 6 If k is any scalar and $\mathbf{u}$ is an object in V , then $k\mathbf{u}$ is in V .

Rule 7 $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

Rule 8 $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

Rule 9 $k(m\mathbf{u}) = (km)\mathbf{u}$

Rule 10 $1\mathbf{u} = \mathbf{u}$

We sometimes write $t \cdot \mathbf{v}$ to emphasize the scaling operation.

Proof exercises for an abstract vector space.

- (1) Show that $\mathbf{v} + \mathbf{v} = 2\mathbf{v}$.
- (2) Show that $\mathbf{v} + \mathbf{v} + \mathbf{v} = 3\mathbf{v}$ (and argue why $\mathbf{v} + \mathbf{v} + \mathbf{v}$ makes sense as an expression).
- (3) Show that if $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$, then $\mathbf{v} = \mathbf{w}$.
- (4) Show that $0 \cdot \mathbf{v} = \vec{0}$. (Note the two different “zeros”.)
- (5) Show that $t \cdot \vec{0} = \vec{0}$.
- (6) Show that $(-1) \cdot \mathbf{v} = -\mathbf{v}$.
- (7) Show that if t is a non-zero real number and $\mathbf{v}$ is a non-zero vector, then $t \cdot \mathbf{v} \neq \vec{0}$.
- (8) Show that if $t \cdot \mathbf{v} = s \cdot \mathbf{v}$, then either $\mathbf{v} = \vec{0}$ or $t = s$.
- (9) Show that if $t \cdot \mathbf{v} = t \cdot \mathbf{w}$, then either $t = 0$ or $\mathbf{v} = \mathbf{w}$.
- (10) Show that if a vector space V has (at least) two distinct elements, then V has infinitely many elements.
- (11) If V and W are vector spaces, show that the set $V \times W$ which consists of all pairs $(\mathbf{v}, \mathbf{w})$ with $\mathbf{v}$ in V and $\mathbf{w}$ in W is also a vector space. What are the operations?

- (12) Suppose V is a vector space, and consider the set $\text{Fib}(V)$ of all Fibonacci-like sequences of the form $\{\mathbf{v}_n\}_{n=1}^{\infty}$ where each $\mathbf{v}_n$ is an element of V and $\mathbf{v}_{n+2} = \mathbf{v}_{n+1} + \mathbf{v}_n$ for all $n \geq 1$. Is $\text{Fib}(V)$ a vector space?
- (13) Find a set which satisfies Rules 1 to 9, but does not satisfy Rule 10.

Exercise to find examples of vector spaces. Do the following sets satisfy the definition of a vector space under the most sensible choices of addition and scalar multiplication?

- (1) $\mathbb{R}^2$
- (2) $\mathbb{R}^3$
- (3) $\mathbb{R}$
- (4) $\mathbb{C}$
- (5) $\mathbb{N}$
- (6) $\mathbb{Q}$
- (7) $\mathbb{Z}$
- (8) A set $X = \{\mathbf{v}\}$ with exactly one element.
- (9) A set

Beatles = {John, Paul, George, Ringo}

with the operations that if t is a real number and $\mathbf{v}, \mathbf{w}$ are elements of Beatles, then
 $t \cdot \mathbf{v} = \text{John}$ and $\mathbf{v} + \mathbf{w} = \text{Ringo}$.

- (10) The set of all sequences of real numbers.
- (11) The set of all bounded sequences of real numbers.
- (12) The set of all increasing sequences of real numbers.
- (13) The set of all “Fibonacci-like” sequences.
 That is sequences of real numbers of the form $\{x_n\}_{n=1}^{\infty}$ where $x_{n+2} = x_{n+1} + x_n$ for all $n \geq 1$.
- (14) The set of polynomials.
- (15) The set of all real-valued functions defined on $\mathbb{R}$.
- (16) The set of all real-valued continuous functions defined on $\mathbb{R}$.
- (17) The set of all real-valued functions defined on $\mathbb{R}$ with the property $f(0) = 7$.
- (18) The set of all real-valued functions defined on $\mathbb{R}$ with the property $f(0) = 0$.
- (19) The set of all real-valued functions defined on $\mathbb{R}$ with the property $f(0) = f(1)$.
- (20) The set of all real-valued differentiable functions defined on $\mathbb{R}$ with the property $f'(0) = 0$.
- (21) The set of all real-valued differentiable functions defined on $\mathbb{R}$ with the property $f'(0) = f(0)$.
- (22) The set of all real-valued differentiable functions defined on $\mathbb{R}$ with the property $f'(0) = f(1)$.
- (23) The set of all Lipschitz functions defined on $\mathbb{R}$.
- (24) The set of all linear functions from $\mathbb{R}^m$ to $\mathbb{R}^n$.